

WASP-10b

R-1434

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_141125 · created 2026-08-01T07:47:34+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456986.678 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.0059
Transit depth (Rp/Rs)^2	3.03 +/- 0.2 [%]
Orbital Inclination (inc)	88.52 +/- 0.61
Airmass coefficient 1 (a1)	1.0175 +/- 0.0078
Airmass coefficient 2 (a2)	-0.0137 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#2 - [408, 196]
Optimal Method	PSF photometry
Transit Duration (day)	0.0929 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.174 ± 0.0059 vs 0.1592 (deviation 2.49σ)
Duration fit vs published	0.0929 d vs 0.0946 d (deviation 0.6σ)
Mid-time O–C	1.5 min
Depth SNR	29.3
Residual scatter	0.76 %

Light curve

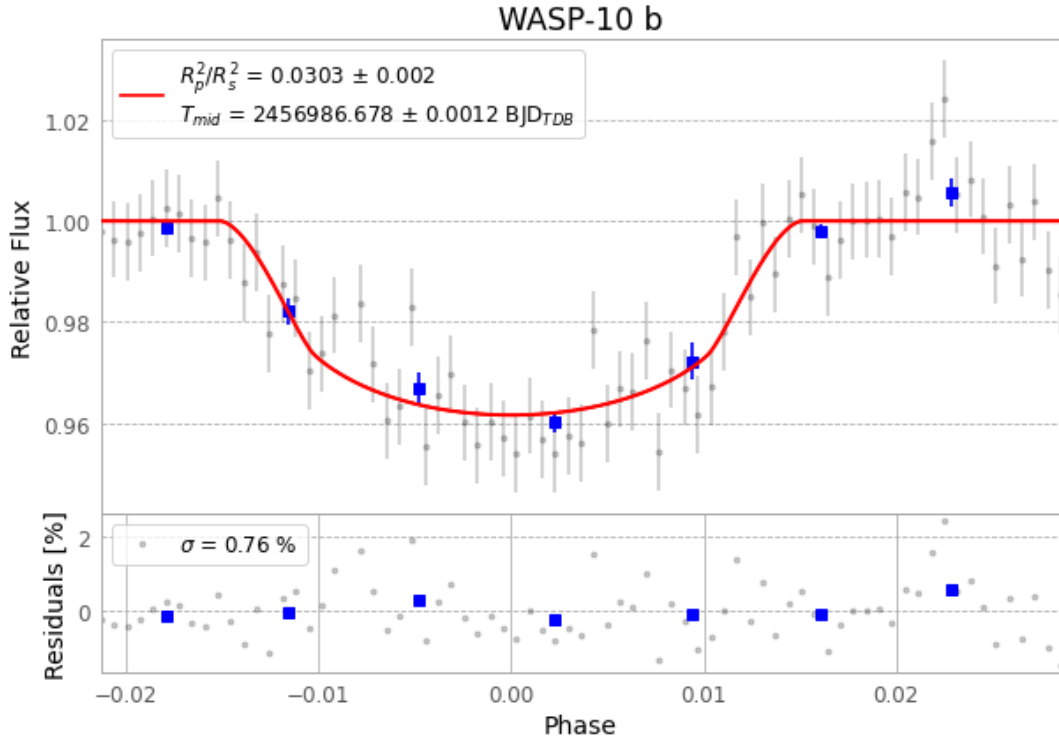


Figure 1 — FinalLightCurve_WASP-10 b_2014-11-24.png (SHA-256 919624565b9fb29e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [466, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d89b794cd35d41ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-10141125023612.FITS → WASP-10141125061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-141125.log (c23e879281d9...), FinalLightCurve_WASP-10 b_2014-11-24.png (919624565b9f...), FinalLightCurve_WASP-10 b_2014-11-24.csv (b13d7e49bd6d...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:47Z.